PRAYAS

UG Level Research Culture — Problem Statement

Problem ID	PRAYAS-INFO-26-S-007
Category	Software
Title	Personalized Student Learning Capability & Comprehension Assessment using Jev and LLM Integration
Domain	Artificial Intelligence & Machine Learning, Natural Language Processing, Web/Mobile Application Development
Faculty Mentor	DEVEN SHAH
Department	Information Technology
Problem Background	Undergraduate students often struggle with patience when dealing with long, complex academic or technical research papers. To bridge this gap, modern tech education requires micro-learning components where students read short, highly applied articles, comprehend them in their own words, and receive automated, objective evaluations of their understanding before the source text disappears.
Problem to be Investigated	How can a combined system of a fast-decision-making model (Jev) and a Large Language Model (LLM) effectively assess and rate student comprehension of technical articles in real-time within an interactive application?
Expected Research Component	Designing a streamlined student evaluation workflow where source text is temporarily displayed and hidden, requiring students to summarize the gist. Implementing and optimizing Jev API endpoints (utilizing null, choice, or score types) alongside an LLM to
Suggested Approach	1. Develop a web or mobile application interface where an article is presented to the student for a designated reading period. 2. Hide the article text and prompt the student to write a comprehension summary/purpose statement in their own words. 3. Send the student's input and source constraints to a backend integrated with an LLM and the Jev model (Typesafe SDK) to perform fast structured evaluations and scoring. 4. Display the accuracy rating (1 to 10) and targeted feedback to the student instantly.
Available Resources	Jev Model Documentation & SDKs (Python/TypeScript) - Video Reference Guide: What is Jev and How to Use it? - Standard LLM API integration tools (OpenAI, Vercel AI Gateway, etc.) - Web/Mobile frontend frameworks (React, Flutter, or Next.js)

Expected Outcome	A fully functional mobile application or web page prototype featuring article display/hide mechanics, student text input handling, and a backend pipeline utilizing Jev and LLM combined logic to generate accurate 1-to-10 comprehension ratings.
Duration	8 Days
Team Size	2 students
Prerequisite Knowledge	Basic Web/Mobile Development, Fundamentals of REST APIs, and basic understanding of LLM prompt engineering and structured data evaluation.
UN SDG Alignment	SDG 4 (Quality Education)
Resource Links	What is Jev and How to Use it? (YouTube)